# How AI Agents Treat Each Other: An Axelrod-Style Iterated Prisoner's Dilemma Tournament with Language-Model Agents A Pilot Study

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As autonomous AI agents begin to act on behalf of people online, they will increasingly interact with one another in repeated encounters where cooperation and exploitation are both possible. We study this setting with an Axelrod-style iterated Prisoner's Dilemma tournament in which large language model (LLM) agents of three sizes compete alongside three fixed-strategy bots. The game includes execution noise (10%), optional non-binding messages, and a public reputation signal. Agents were assigned one of three goal framings: neutral (maximise total points), cooperative (maximise points while valuing fairness), or competitive ("beat your opponents" in each match). Across 20 tournaments in four settings, neutral and cooperative agents almost never defected first and forgave most noise-induced defections. Under the competitive framing we observed two different strategies: a cooperate-then-betray strategy accompanied by friendly messages, which won its setting, and unconditional defection, which won most head-to-head matches yet finished 7th or 8th of 8, consistent with Axelrod's observation that envious strategies lose. Within a setting, the same model behaved very differently under neutral and competitive goals, while neutral agents of three sizes behaved similarly; however, the design does not isolate the relative effects of goal framing and model size. No agent was persuaded by a lying bot, and an unconditionally defecting player coincided with lower scores for the other players. A follow-up check showed that the competitive agent's opening move is close to a coin flip and that batching several games into one model call makes games non-independent. With five runs per setting and no statistical tests, these results are exploratory. We state explicitly which claims the pilot supports and propose a controlled factorial follow-up.

## Introduction

AI agents are beginning to act for people on the internet: buying, selling, negotiating and exchanging information. In many of these interactions, the counterpart will itself be an AI agent, and the same agents will meet repeatedly. Whether such agents cooperate or exploit one another will shape the norms of this emerging "agent internet".

The iterated Prisoner's Dilemma (IPD) is the canonical model of this tension. In a single round, defection dominates; yet when the game is repeated, conditional cooperation can be sustained. In his seminal computer tournaments, Axelrod  found that the simple strategy *Tit-for-Tat* (cooperate first, then copy the opponent's previous move) performed best. He summarised successful strategies as *nice* (never defect first), *retaliatory*, *forgiving*, and *not envious*: a strategy that tries to beat each opponent, rather than to do well itself, tends to do poorly overall .

This study replaces hand-coded strategies with LLM agents that read the game state in natural language and decide each move. We address four research questions:

1. Do LLM agents cooperate with one another, or exploit one another?
2. Does the wording of an agent's goal change its behaviour?
3. Does model size (small, medium, large) matter?
4. Do non-binding messages (cheap talk) support cooperation or deception?

# Method

## Game

Each tournament had eight players in a round-robin format: every player played a separate 10-round match against each of the other seven. Players were told that the tournament could end after any round and were not told its length. Payoffs followed Axelrod's standard values (Table 1).

*Payoff matrix (row player's points, column player's points).*

| | Opponent: C | Opponent: D |
|---|---|---|
| Player: C | (3, 3) | (0, 5) |
| Player: D | (5, 0) | (1, 1) |

Three further features made the environment closer to real agent interactions:

- **Noise.** Each move was flipped with probability 0.10. Players saw their own intended and actual moves, but only their opponent's *actual* moves, so accidental defections could not be told apart from deliberate ones.
- **Messages.** With each move, a player could send a message of at most 25 words, delivered at the start of the next round. Messages were non-binding and could be false.
- **Reputation.** Each player saw, for every opponent, the share of cooperative moves that opponent had played against *other* players.

Player identities were hidden behind letters A–H, shuffled in every run.

## Players

Each tournament contained five LLM agents and three fixed-strategy bots (Table [tab:players]). The small, medium and large agents were Claude Haiku, Sonnet and Opus models, respectively.

@lL@ Player & Goal or rule
Small, medium, large AI & *Neutral*: "earn as many total points as possible, summed across

all of your matches."
Cooperative AI & Neutral goal, plus: "you personally value fairness and mutual benefit."
Competitive AI & "BEAT your opponents. In each match, try to finish with MORE points than that opponent."
Tit-for-Tat bot & Cooperates first, then copies the opponent's previous actual move.
Random bot & Plays C or D with equal probability.
Cheat bot & Always defects. When messages are enabled it sends friendly lies (e.g. "I always cooperate").
Grudger bot & Cooperates until the opponent's first defection, then always defects (B2 only).

## Settings

Four settings were run, five tournaments each (Table 2). B1 and B2 reused the random seeds of S1, so seat assignments and noise draws were paired with S1.

*Experimental settings (5 runs each).*

| Setting | Competitive AI | Cooperative AI | Messages | Third bot |
|---|---|---|---|---|
| S1 | Medium | Medium | On | Lying cheat bot |
| S2 | Large | Small | On | Lying cheat bot |
| B1 | Medium | Medium | Off | Silent cheat bot |
| B2 | Medium | Medium | Off | Grudger bot |

## Procedure

In each round, every AI agent received a written prompt containing the rules, its goal, the full history of each of its matches, opponents' reputations and the latest messages. It returned a structured answer with one move per opponent, optional messages, and a brief rationale. Bot moves, noise and scores were computed by a game engine. All reported scores were recomputed from the raw move logs.

In a pilot run and in rounds 1–3 of S1, each agent was queried separately for each game. To reduce cost, later rounds used a single call per agent role that played the same seat in all parallel games of the round (5 games in S1 and S2; 10 in B1 and B2). Section 4 shows that this design choice matters.

## Measures

- **Score:** total points over all matches, averaged over runs.
- **Cooperation rate:** share of intended moves that were C.

- **Unprovoked betrayal:** an intended D against an opponent whose actual moves in that match had all been C (matches against the cheat bot excluded).
- **Forgiveness:** share of cases in which, after an opponent's intended C was flipped to D by noise, the player chose C in the next round.
- **Head-to-head wins:** number of matches (of 7) in which the player outscored its opponent.

# Results

## Scores

Figure [fig:scores] shows average scores for every player in each setting; full numbers are in Appendix 10. The competitive AI ranked first only in S1. In S2, B1 and B2 it finished 7th or 8th of 8, while Tit-for-Tat finished first or second in every setting.

## Behaviour

Table [tab:behaviour] summarises behaviour across all 20 tournaments.

@lccL@ Player & Cooperation rate & Unprovoked betrayals & Forgiveness of noise
Cooperative AI & 67–82% & 2 & 87% (83/95)
Large AI & 67–79% & 0 & 79% (67/85)
Small AI & 66–76% & 7 & 68% (52/77)
Medium AI & 66–80% & 0 & 64% (38/59)
Tit-for-Tat bot & 63–71% & 0 & 0% (0/107)
Competitive AI & 53% (S1); 0% (S2, B1, B2) & 20 (S1) & 33% in S1 (11/33); defected every round elsewhere

The main patterns were:

1. **Two competitive strategies were observed.** In S1, the medium competitive agent cooperated about half the time, committed 20 unprovoked betrayals (19 of them alongside cooperative-sounding messages), and won 3 of 5 tournaments. In S2, the large competitive agent defected from round 1 and sent no messages, stating that it preferred not to make promises it would not keep. Because model size, cooperative-agent size and the setting all differed between S1 and S2, and because the follow-up check (Section 4) found the opening move to be close to random, this contrast cannot be attributed to model size.
2. **Unconditional defection scored poorly.** Whenever the competitive agent defected unconditionally (S2, B1, B2), it won 5.4–5.6 of its 7 matches on average yet scored lowest among all AI agents, because mutual defection yields 1 point per round instead of 3.

3. **Nice and forgiving majority.** Neutral and cooperative agents almost never defected first (9 cases in 20 tournaments) and forgave 64–87% of noise-induced defections, compared with 0% for Tit-for-Tat.

4. **Actions over words.** After round 1, agents cooperated with the lying cheat bot on average at most 0.8 times in 9 rounds, despite its friendly messages.

5. **Lower scores alongside a defector.** With an unconditional defector present (S2, B1), the other AI agents scored roughly 10–20 points less than in S1. Other factors also changed between these settings, so this is an association, not a measured causal effect.

6. **Goal framing and model size.** Neutral agents of all three sizes scored within 11 points of each other in every setting, while the same medium model behaved very differently under neutral and competitive goals within S1. The competitive goal was never given to the small model, and was given to the medium and large models only in different settings, so the design cannot compare the size of the two effects or test their interaction.

## Follow-up Check: Opening Move and Call Batching

In B1 and B2 the medium competitive agent defected in every round of all 10 runs, suggesting that without messages it abandons the cooperate-then-betray strategy. However, in these settings a single model call decided round 1 for all 10 games and produced identical moves in each. The 10 runs therefore reflected one decision, not ten.

To separate the effect of messages from the effect of batching, we queried the medium competitive agent six times on a single round-1 game, each time in a fresh call. The prompts were identical except for the message rules (Table 3).

*Opening moves of the medium competitive agent in fresh single-game calls.*

| Prompt | Opening moves | Opened with C |
|---|---|---|
| With messages (S1) | C, D, C | 2 of 3 |
| Without messages (B) | D, C, D | 1 of 3 |

All defecting openings gave the same rationale: defection can never leave the agent behind its opponent in a single round. All cooperating openings cited probing for reciprocity. The agent appears to choose between these two lines of reasoning almost at random. Messages may raise the probability of a cooperative opening, but three calls per condition cannot establish this. Methodologically, batching several games into one call makes those games non-independent and overstates the effective sample size.

## Discussion

*Goal wording may be a key lever.*

Within S1, a single sentence of goal framing changed the same model from a reliable partner into an exploiter. If this holds in controlled studies, the behaviour of multi-agent systems may depend substantially on how deployers phrase objectives, and evaluations should test sensitivity to goal wording.

*Axelrod's principles appear to transfer to LLM agents.*

Nice, forgiving and non-envious behaviour performed well. When the relative-payoff (envious) objective led to unconditional defection, it won matches but lost tournaments; when it led to cooperate-then-betray (S1), it won. A one-line Tit-for-Tat rule was the most consistently high-ranking player.

*Forgiveness is exploitable.*

High forgiveness keeps cooperation alive under noise, but a strategic agent can disguise deliberate defections as accidents. Robust agents may need to remain forgiving while tracking patterns of repeated "accidents".

*Cheap talk is double-edged.*

Messages alone did not sustain deception: no agent kept trusting the lying bot, whose actions and reputation contradicted its words. Yet messages accompanied the most successful exploitation in S1. Reputation and verifiable track records may matter more than promises in agent ecosystems.

*Aggressive agents may impose costs on everyone.*

Settings with an unconditional defector showed lower scores for the other players. This is expected from the payoff structure, but it was not isolated experimentally here; if confirmed, a small number of aggressive agents could degrade outcomes in open multi-agent networks.

## What This Pilot Can and Cannot Claim

Table [tab:claims] classifies the main statements of this paper by the strength of the evidence behind them.

@LlL@ Claim & Status & Reason
Neutral and cooperative agents rarely defected first and often forgave noise & Supported (descriptive) & Consistent in all 20 tournaments; counts reported directly from logs
No agent kept cooperating with the lying bot & Supported (descriptive) & Consistent across all settings with that bot
Unconditional competitive defection won matches but scored poorly overall & Supported (descriptive) & Seen in S2, B1 and B2; follows from the payoff structure

The same model behaves differently under neutral and competitive goals & Suggestive & Shown within S1, but only for one model and 5 runs, without statistical tests
Model size has little effect under a neutral goal & Suggestive & Small score differences, but 5 runs per setting and no tests
Removing messages changes the competitive agent's strategy & Not established & The B result reflects one batched decision; the fresh-call check (3 vs 3) is inconclusive
Goal wording matters more than model size & Not established & Goal and size were not varied independently; no interaction test
Larger models defect more under a competitive goal & Not established & S1 and S2 differ in several variables, and the opening move is close to random
A defector causes lower scores for others & Not established (causally) & Observed association; settings differ in several variables

## Limitations

- **Sample size and statistics.** Five runs per setting provide early signals, not statistical evidence; no significance tests were run. The required sample size should be set in advance by a power analysis.
- **Confounded settings.** Settings differed in more than one variable at a time (model size of the competitive and cooperative agents, messages, third bot), so differences between settings cannot be attributed to a single cause.
- **Mixed querying method.** Per-game and batched calls were mixed across rounds and settings; batching introduces dependence between games.
- **Single model family.** All agents were Claude models; other model families may behave differently.
- **Short horizon and fixed prompts.** Matches lasted 10 rounds, and only one wording of each goal and of the rules was tested.
- **One information leak.** In one round, an agent saw other players' answer files after submitting its own; instructions were tightened afterwards.

## Future Work

The most informative next step is a controlled factorial experiment on the opening decision: goal framing (neutral, competitive) × messages (enabled, disabled) × model size (small, medium, large), giving 12 conditions. All other variables would be held fixed, every observation would come from an independent single-game call, the primary measure (probability of a cooperative opening) would be defined in advance, and the sample size per condition would be chosen by power analysis. Beyond this, we plan: (i) a clean replication of S1, S2 and B with per-game calls and 20–30 runs per setting; (ii) a control setting without a competitive agent; (iii) an all-AI society with one competitive agent; (iv) a

goal-wording study comparing “win overall”, “beat each opponent” and “earn the most” across model sizes; (v) an evolutionary variant in which low scorers adopt the top scorer’s goal over several generations; and (vi) comparison with other model families, including open-weight models.

## Conclusion

In an Axelrod-style tournament, LLM agents with neutral or cooperative goals behaved much like Axelrod’s successful strategies: nice, forgiving and resistant to deceptive talk. Under a competitive goal framing we observed either successful deceptive exploitation or self-defeating unconditional defection. The pilot suggests that goal framing strongly shapes agent behaviour, but its design does not isolate goal framing from model size, messaging or chance. These exploratory results motivate a controlled, adequately powered factorial study of how AI agents will treat each other.

## AI Assistance Statement

The research idea, questions and study design are the author’s. AI tools (Claude, by Anthropic) were used to implement the game software, run the simulations, analyse the logs and draft this manuscript. The AI players in the tournaments were also Claude models, as described in Section 2. All game logs, exact prompts, model responses, the follow-up check and the simulation and analysis code are available from the author on request, so that every reported number can be independently recomputed.

3 R. Axelrod. Effective choice in the Prisoner’s Dilemma. *Journal of Conflict Resolution*, 24(1):3–25, 1980.

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## Full Score Table

*Average points per player (5 runs per setting); rank in parentheses. Scores were recomputed from raw move logs.*

| Player | S1 | S2 | B1 | B2 |
|---|---|---|---|---|
| Competitive AI | **181.6** (1) | 141.8 (7) | 140.2 (7) | 144.8 (8) |
| Tit-for-Tat bot | 171.4 (2) | **159.8** (1) | **160.0** (1) | **173.0** (1) |
| Large AI | 169.8 (3) | 156.8 (2) | 158.4 (2) | 163.4 (3) |
| Small AI | 169.0 (4) | 154.6 (3) | 152.6 (4) | 159.0 (5) |

| Player | S1 | S2 | B1 | B2 |
|---|---|---|---|---|
| Medium AI | 168.2 (5) | 148.6 (5) | 147.4 (5) | 158.4 (6) |
| Cooperative AI | 168.2 (5) | 154.6 (3) | 153.6 (3) | 162.4 (4) |
| Random bot | 144.8 (7) | 148.2 (6) | 144.0 (6) | 150.2 (7) |
| Cheat bot | 132.8 (8) | 131.2 (8) | 133.2 (8) | — |
| Grudger bot | — | — | — | 169.8 (2) |

In S2 the competitive AI was the large model and the cooperative AI the small model; in all other settings both were the medium model.